Chenyu Wang

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EDUCATION

University of Michigan (GPA: 4.0/4.0) Ann Arbor, MI
Electrical & Comp Engin MEng, Data Sci & Machine Learning. ***Sep 2021 – Dec 2022***
Hangzhou Dianzi University (GPA: 4.05/5.0 | Ranking: 1/23) Hangzhou, China
B.E. in Information Engineering. ***Sep 2017 – June 2021***

RESEARCH EXPERIENCE

University of Michigan ***Research Assistant / Advisor: Prof. Hun-Seok Kim***

Deep Learning-Based Joint Channel Coding and Frequency Modulation *Jun 2022 – Present*

- Build a novel RNN(GRU)-based Joint trained encoder and decoder with training methodologies.
- Leveraged a hyperparameter optimization tool Optuna to jointly optimize channel autoencoder.
- Conducted local permutation method to explain the coding strategy of GRU-based encoder.

Hangzhou Dianzi University & Shanghai Jiao Tong University ***Research Assistant / Advisor: Prof. Meihua Bi***

Research on Simplified SVM equalization algorithm based on single hyperplane training. *Apr 2021 – Aug 2021*

- Performed complexity analysis for the machine learning algorithm involved.

Interpretable framework for ML-based equalizers in IM/DD systems. *Dec 2020 – Mar 2021*

- Applied the SHAP (SHapley Additive exPlanations) method for improving the interpretability of equalizers, constructed an interpretable and visualized SHAP framework for ML/NN-based equalization.
- Reduced half of the taps/features used for training ML-based equalizer without deteriorating system performance verified in the 50Gb/s PAM-4 per wavelength IM/DD system with 10GHz optical devices.

LightGBM based Equalization Scheme for High-Speed Optical Transmission. *Dec 2019 – Nov 2021*

- Constructed a LightGBM based equalizer to compensate transmission impairments of 25 Gb/s NRZ and 50 Gb/s PAM-4 modulated signals.
- Conducted feature selection with feature importance calculated by statistical data *gain* and *split* extracted from trained LightGBM booster.

PUBLICATIONS

- Chang, B., **Wang, C**., & Kim, H. S. *Deep Learning-Based Joint Channel Coding and Frequency Modulation for Low Power Connectivity*. Accepted by International Conference on Communications**(ICC), IEEE, 2023**.
- **Wang, C**., Miao, X., Xi, Y., Bi, M., Li, L., & Hu, W. *Interpretable and visualized SHAP-based equalizer with feature selection in IMDD system.* In Optical Fiber Communications Conference and Exhibition**(OFC)**, **IEEE, 2021**.
- Xi, Y., Bi, M., Xin, M., **Wang, C**., Yu, J., & Hu, W. (2021). Simplified SVM Equalization Algorithm Based on Single Hyperplane Training Enabled 50Gb/s PAM-4/8 With 10-G Optics in NG-PON System. **IEEE Photonics Journal**, **2021**.
- **Wang, C**., Miao, X., Xi, Y., Bi, M., & Hu, W. A New LightGBM-Based Equalizer Enabled High-Capacity PAM-4 and NRZ Transmission in the 10-G Class System. In (ACP/IPOC), 2020.
- Xi, Y., **Wang, C**., Miao, X., Bi, M., & Hu, W. High-Efficient Equalizer Based on the Simplified Deep Neural Network for 56Gb/s/λ PAM-4 in C-band 10G DML-Based Short Reach System. In (ACP/IPOC), 2020.

PROJECTS EXPERIENCE

EECS 605 Data Science and Machine Learning Design Laboratory *Feb 2022 – Apr 2022*

Fast image style transfer Web App deployed on Heroku.

- Developed a robust cloud pipeline in Amazon AWS cloud for image style transformation.
- Deployed a React web app in Heroku with Rest APIs for real-time image style transfer.

EECS 545 Machine Learning *Oct 2021 – Dec 2022*

Implementation and Interpretation of XGBoost

- Designed, implemented and validated a working version of XGBoost and its three built-in methods for calculating the feature importance in Python from the scratch.
- Explored the improvement and interpretability between different built-in feature selection methods of XGBoost.